

1. (a) $\Omega = \{(C_a, C_a), (C_a, C_o), (C_o, C_a), (C_o, C_o)\}$ ou $\Omega = \{(a_1, a_2) : a_1, a_2 \in \{C_a, C_o\}\}$ ou $\Omega = \{C_a, C_o\} \times \{C_a, C_o\}$;
 (b) $\Omega = \{(a_1, a_2) : a_1 \in \{C_a, C_o\}, a_2 \in \{1, 2, 3, 4, 5, 6\}\}$ ou $\Omega = \{C_a, C_o\} \times \{1, 2, 3, 4, 5, 6\}$;
 (c) $\Omega = \{(a_1, a_2, a_3) : a_1, a_2, a_3 \in \{1, 2, 3, 4, 5, 6\}\}$ ou $\Omega = \{1, 2, 3, 4, 5, 6\}^3$
2. a) $\Omega = \{1, 2, 3, 4, 5, 6\}$ e $P(\{i\}) = \frac{2^{i-1}}{63}$, $i \in \{1, \dots, 6\}$;
 b) Não;
 c) $A = \{2, 4, 6\}, P(A) = \frac{42}{63}$; $B = \{3, 6\}, P(B) = \frac{36}{63}$; $\overline{B} = \{1, 2, 4, 5\}, P(\overline{B}) = \frac{27}{63}$;
 $C = \{1, 3, 5\}, P(C) = \frac{21}{63}$; $A \cap B = \{6\}, P(A \cap B) = \frac{32}{63}$;
 $A \cup B = \{2, 3, 4, 6\}, P(A \cup B) = \frac{46}{63}$; $A \setminus B = \{2, 4\}, P(A \setminus B) = \frac{10}{63}$;
 $A \cup B \cup C = \Omega, P(A \cup B \cup C) = 1$.
3. a) $\emptyset$; $\emptyset$; Ω ; Ω ;
 b) $D = A \cap B$; $E = B \cap C$; $F = A \cup B \cup C$; $G = \overline{A} \cap \overline{B} \cap \overline{C} = \overline{F}$;
 $H = (A \cap \overline{B} \cap \overline{C}) \cup (\overline{A} \cap B \cap \overline{C}) \cup (\overline{A} \cap \overline{B} \cap C)$
4. (a) 0.32; (b) 0.68; (c) 0.12; (d) 0.24
5. (a) 0.485; (b) 0.515; (c) 0.025; (d) 0.165; (e) 0.035; (f) 0.295
6. (a) 0.2; (b) 0.575; (c) 0.044
7. (a) 0.72; (b) $\frac{0.1}{0.28}$; (c) Não
8. Sim; Sim; Não; Não
9. (a) Não; (b) Falsa
10. (a) 0.1; 0.6; 0.3; 0.36; 0.42; (b) $\frac{0.52}{0.64}$

Exercícios Suplementares à Folha 1

1. (a) $\frac{260}{630}$; (b) $\frac{120}{630}$; (c) $\frac{470}{630}$; (d) $\frac{160}{630}$; (e) $\frac{380}{630}$
2. $P(A \Delta B) = P(A) + P(B) - 2P(A \cap B)$
3. (a) 0.32; (b) 0.68; (c) 0.03; (d) 0.09; (e) 0.03; (f) 0.21
4. (a) $\frac{30}{36}$; (b) $\frac{6}{30}$; (c) $\frac{24}{30}$; (d) $\frac{3}{6}$; (e) Não
5. $\frac{2 \times \binom{13}{5}}{\binom{26}{5}} = \frac{9}{230}$
6. (a) 0.47; 0.55; 0.85; 0.83; (b) $\frac{0.38}{0.55}$; (c) $\frac{0.17}{0.55}$; (d) $\frac{0.15}{0.45}$; (e) $\frac{0.38}{0.53}$
7. (a) $\Omega = \{C_a, C_o\}^{n-1}$;
 (b) i. $P(E_j) = \frac{1}{2}, j \in \{1, \dots, n\}$; ii. $P(\emptyset) = 0$ e não são independentes
8. $n > 6.644$ pelo que $n = 7$.
9. Se $n = 2$ não são independentes; se $n = 3$ são independentes.
10. (a) — (b) 0.719; $\frac{0.665}{0.719}$; $\frac{0.246}{0.281}$